LIANG-JIE CHIU

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SUMMARY

Data scientist seeking to bring ML/DL expertise to baseball operations, with applied work in counterfactual swing-decision valuation, transformer-based pitch outcome prediction, MCTS pitch sequencing, and survivorship-corrected aging curves.

EDUCATION

Columbia University Expected Dec 2026
MS in Data Science, GPA: 3.83/4.0 New York, NY

National Sun Yat-sen University (NSYSU) Sep 2020 - Jun 2024
BBA in Information Management, GPA: 3.97/4.0 Kaohsiung, Taiwan

PROFESSIONAL EXPERIENCE

ASTA Health Tech Jun 2026 - Aug 2026
AI/ML Research Intern Jersey City, NJ (Remote)

- Implemented Gaussian (mu, sigma) head for 30-minute multi-step vital-sign forecaster, delivering calibrated 50/80/90% uncertainty bands and beating persistence by 28.5% on 1.19M-window dataset rebuilt from PostgreSQL dumps.
- Architected ECG vision pipeline on photographed monitors, pairing D-FINE region detector (0.909 macro-F1 vs 0.617 fixed-box baseline) with 1D-heatmap ResNet18 beat localizer (0.554 vs 0.414 periodic baseline).

TIFIN Feb 2026 - May 2026
Data Science Intern (Capstone Project) New York, NY (Remote)

- Designed three-layer Reasoner-Answerer-Verifier pipeline grounding financial explanations in API outputs, with Verifier enforcing 6 factual conditions and triggering corrective retries, evaluated across 227 sessions on 5 LLM-judge metrics.

Research Center for Information Technology Innovation, Academia Sinica Jul 2024 - Jul 2025
Full-Time Research Assistant Taipei, Taiwan

- Increased liver tumor Dice score by 11% over UNet++ with weighted combined loss and HU windowing for CT preprocessing, trained on 80 GB of CT data across LiTS17 and HCC-TACE-Seg.
- Developed two-stage federated YOLOv8 and TabNet system for orchid double-spike prediction, raising small-farm AUC from 0.69 to 0.82 through federated training and continual-learning pipeline.

BASEBALL PROJECTS

MLB Swing Decision Value Fall 2026

- Modeled counterfactual swing/take run values on 4.28M Statcast pitches with XGBoost whiff/foul/in-play and batted-ball-event classifiers (held-out whiff AUC 0.77), re-projecting pre-2026 pitch locations to ABS-era frame.
- Benchmarked model variants by paired held-out MSE with game-resampled 95% CIs; shrank hitter-specific adjustments to 68% strength, lifting calibration slope from 0.95 to 1.01; shipped Streamlit dashboard ranking 434 hitters.

MLB Pitch Outcome Prediction and Sequence Optimization Spring 2026

- Built multi-task transformer predicting pitch outcomes, hit locations, and batted-ball physics via hierarchical pitcher-batter encoder with cross-attention, outperforming LightGBM on 2025 Statcast data (top-4 recall 0.963/0.691 vs. 0.958/0.642).
- Engineered expectimax MCTS pitch sequencer using trained transformer as simulator, analytically averaging outcome distributions over RE288 run-expectancy rewards with exact Bellman backup to rank pitch-zone actions.

MLB Position Player Aging Analysis Spring 2026

- Mitigated survivorship bias in GAM aging curves by fitting per-metric survival models (age, playing time, experience, lagged performance) as IPW weights, cutting test MAE 7.3% (wRC+) and 4.3% (WAR) over a delta-method baseline.
- Quantified wRC+ peak age at 27 across 45 MLB seasons, validated by refitting on random player halves (peak stable within 0.6 years), with elite hitters (career wRC+ $\geq$ 110) peaking a year later at 28, informing long-term contract structure.

TECHNICAL SKILLS

Programming & Data Tools: Python, R, SQL, C/C++, Spark, GCP, BigQuery, Airflow, pandas, polars.
Machine Learning: PyTorch, HuggingFace Transformers, PEFT/LoRA, Scikit-learn, pybaseball, statsmodels.